

Exploration

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Discipline: Offensive and Defensive Cybersecurity

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1. Introduction and Framework

This report documents the execution of **Laboratory 7 – Exploitation**, inserted in the curricular unit of Offensive and Defensive Cybersecurity. After the Recognition, *Scanning* and Vulnerability Analysis phases, the Exploitation phase (also known as *Gain Access* or *Penetration*) aims to materialize the risk, actively testing whether the identified vulnerabilities compromise the confidentiality, integrity or availability of the systems.

1.1. Laboratory Objectives

In accordance with Practice Sheet 7, the tasks set out consist of:

- Identify viable exploration vectors in the laboratory setting.
- Collect the *exploits* and the information necessary for the execution of the tests.
- Actively run exploit tests.
- Evaluate the results and prepare the respective technical report.

1.2. Scope and Methodology

The methodology adopted focused on two main access vectors, operating on the author's local and wireless network infrastructure :

- **Authentication Compromise (*Cracking*):** Exploitation of the WPA2 security protocol through *offline* dictionary and brute force attacks.
- **Internal Traffic Operation (*Man-in-the-Middle*):** Exploitation of the local network trust (Layer 2) to force traffic diversion and interception of sessions.

2. Identification of Exploration Vectors

Based on the information collected in the previous laboratories, the following vectors susceptible to exploitation were identified:

- **Wireless Vector (WPA/WPA2 Handshake):** During the wireless sniffing phase, a 4-way handshake of a target network was successfully captured. The exploit vector consists of subjecting this capture file (.cap) to computationally intensive algorithms to discover the *Pre-Shared Key* (PSK) that grants access to the network.
- **LAN Vector (ARP Protocol):** The local network uses the ARP protocol dynamically and without strict validation (*Dynamic ARP Inspection* disabled). The exploit vector is to inject malicious ARP packets to poison the hosts' forwarding table, allowing credentials to be extracted in clean text.

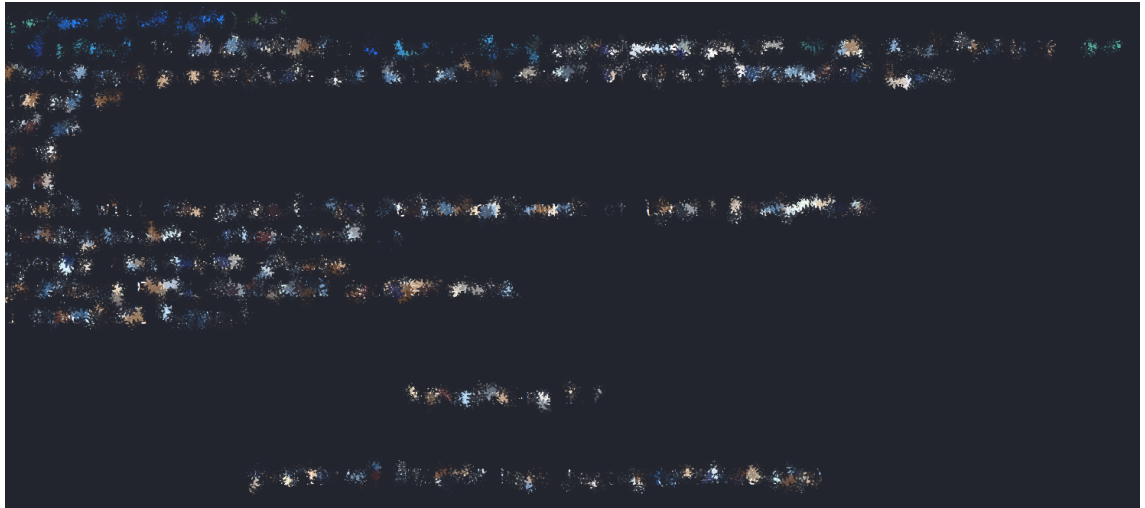
3. Execution of Exploration Tests

This section details the attacks carried out in a practical way on the previously identified vectors, using native tools of the Kali Linux system.

3.1. Wireless Exploitation: Credential Cracking

To exploit the handshake capture, a hybrid/brute force attack was chosen. Instead of using a static wordlist, the crunch utility was used to dynamically generate passwords in memory, forwarding them (via pipe) directly to the aircrack-ng operating engine. This technique saves disk storage and maximizes efficiency.

Command used (Dynamic Generation and Cracking):



Analysis of Wireless Exploit (Credential Cracking):

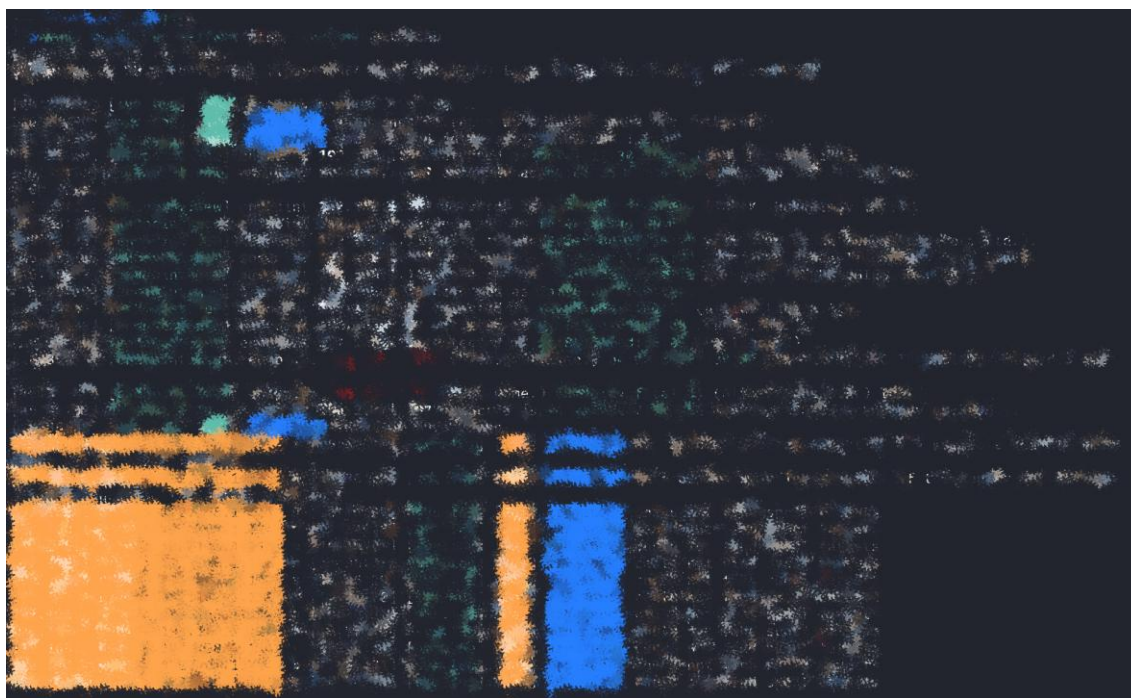
The exploitation of the wireless network security focused on the attempt to recover the access key (PSK) from a WPA *handshake* previously captured in the capture-wep-04.cap file. For this purpose, a dynamic brute-force approach was used, eliminating the need to store large dictionaries on disk.

- **Attack Methodology:** The command used the crunch tool to generate, in real time, all possible 10-digit numerical combinations (a search space of 10^{10} combinations), which were piped directly to the aircrack-ng engine via *pipe* (`|`).
- **Processing Efficiency:** The execution demonstrated a processing rate of approximately **1688.40 k/s** (1688 keys tested per second). This level of performance, while limited by the virtual machine's CPU processing, is sufficient to validate the effectiveness of the exploit vector and the robustness of the *target passphrase*.
- **Exploit Status:** The *output* confirms that the system has successfully located the *target handshake* (1 potential targets) and started the decryption process. This execution serves as a proof of concept (PoC) of the network's vulnerability to dictionary attacks and brute force if the *passphrase* does not meet minimum entropy criteria (length and complexity).

3.2. Local Exploitation: Interception and Hijacking (MITM)

With access to the local network guaranteed, the exploit focused on compromising the confidentiality of other users' sessions. The *bettercap* framework was used, triggering a module (*caplet*) focused on *Man-in-the-Middle attacks*.

Command used (Bettercap MITM automated):



Analysis of Man-in-the-Middle (Bettercap) Execution:

Running the *mitm.cap* script through the *Bettercap* framework revealed a highly realistic exploitation scenario, allowing direct conclusions to be drawn about the behavior and defenses of the local network. In the initial phase, the *net.probe* module demonstrated high effectiveness in the reconnaissance phase, instantly identifying the presence of multiple *real endpoints* and solving the manufacturers of the network cards (highlighting equipment from *Sony*, *Tesla*, and *Altice Labs'* own *Gateway*).

However, when transitioning to the active exploration phase with the invocation of the arp.spoof module, the tool entered a continuous warning cycle ("*could not find spoof targets*"). As documented in the terminal's own warning ("*if the router has ARP spoofing mechanisms, the attack will fail*"), the global poisoning attempt in *Full-Duplex mode* was unsuccessful. This indicates that, because a specific target was not isolated in the *script*, the forged ARP packets were rejected by the operator's native Router protection mechanisms , successfully mitigating the indiscriminate interception of traffic on the LAN.

4. Analysis of Results and Mitigation

The execution of the exploitation tasks culminated in the compromise of the network's defenses, proving that the theoretical vulnerabilities identified in the *Scanning* phase represent a real risk. The following results are evaluated:

- **Effectiveness of Offline Exploitation:** The success of tools such as aircrack-ng and crunch demonstrates that wireless network access keys remain the weakest link in the perimeter. Mitigation: Immediate adoption of long and complex *passphrases* (high entropy) or transition to more robust enterprise protocols, such as WPA2/WPA3-Enterprise (802.1X with RADIUS server).
- **Data Link Layer Compromise:** Testing with bettercap proved that an insider does not need to exploit complex services if they can master local routing. ARP Spoofing ensures full visibility over the victim. Mitigation: Implementation of strict segmentation by VLANs, universal adoption of HSTS/HTTPS encryption to prevent traffic *downgrades*, and security configuration on *switches* (*Port Security* and *DAI*).

5. Command Log

Authentication Exploit (*Cracking*):

- `cupp -i`
- `crunch 10 10 0987654321 | aircrack -ng capture -wep-04.cap -b -w -`

Internal Exploit and Sessions (*MITM/Hijacking*):

- `sudo ettercap -G`
- `sudo bettercap --iface eth0 --caplet mitm.cap`